

Vitthal Bhandari

Seattle, WA • vitthalbhandari2027@gmail.com • vitthal-bhandari.github.io

EDUCATION

University of Washington, Seattle, USA

Expected Aug 2027

MS in Computational Linguistics / GPA: 3.78

Coursework: Statistical Techniques for NLP, Deep and Shallow Techniques for NLP, Speech for Endangered Languages

Birla Institute of Technology and Science Pilani, India

July 2021

BE in Computer Science, Minor in Data Science / GPA: 8.32

Coursework: Probability and Statistics, Applied Statistical Methods, Optimization, Foundations of Data Science

RESEARCH EXPERIENCE

Automatic Speech Recognition for 21 Endangered Languages

[GitHub Repo](#)

- Fine-tuned MMS-1B and XLS-R-1B models with parameter-efficient attention adapters and fully fine-tuned Whisper-large-v3 on a dataset of 21 low-resource languages from Mozilla Common Voice (PyTorch, Hugging Face)
- Implemented n-gram language model decoding through KenLM pipeline and beam search, leading to 27% WER reduction
- Conducted data curation experiments using a char-level 4-gram model to score training utterances by perplexity; MMS fine-tuned on the bottom-60% perplexity subset converged faster to the same WER for 3 languages

Language Modeling from Scratch

[GitHub Repo](#) / [Blog](#)

- Implemented a BPE tokenizer, transformer block, attention architecture, and LM from scratch (PyTorch, Python); used heap and parallelization (multiprocessing) to reduce tokenization time on 2.3GB Tiny Stories dataset by 72%

Detecting Homophobia and Transphobia in YouTube Comments

[GitHub Repo](#) / [Paper](#)

- Fine-tuned pretrained mono/multilingual models (Hugging Face, Python) at the 2nd [LT-EDI Workshop](#) shared task for hate speech detection; implemented easy-data-augmentation (EDA) to tackle class imbalance; ranking 3rd / 6th / 9th on code-mixed / Tamil / English tracks (macro-F1: 0.54 / 0.72 / 0.46)

WORK EXPERIENCE

American Express – Software Engineer II

Mar 2023 – Oct 2025

- Developed a scalable, nested file storage platform (TypeScript, React, Zustand) for legal/tax documents, serving 500+ enterprise users; reduced search and retrieval time from 60 sec to 2 sec and replaced legacy DB handling 2 TB of files
- Implemented a language translation feature using MarianMT model (Hugging Face, Python) for translating English and Spanish chats, with a BLEU score of 0.48 on an internal test set, and serving 8 translations/day for 50+ enterprise users
- Created an automatic dependency manager (Next.js, Sanic, TypeScript) to highlight and resolve package conflicts and vulnerabilities in Node and Python projects (Presented at Amex's global hackathons: GrowthHack 2023, DevCon 2024)

Standard Chartered Bank – Software Engineer I

July 2021 – Feb 2023

- Led a team of 4 in developing an API management framework (TypeScript, React, Flask) for creating configurable APIs (1.3K+ APIs) to fetch/add/update data from Oracle SQL and Dremio, reducing API deployment time from 48 hr to 2 min
- Built reusable UI components (JavaScript, React, Redux) and REST APIs (Flask) for the in-house chatbot [Maya](#), improving the look and flow, while enhancing its capability to analyze data with speed and scale

PayPal – Software Engineer Intern

Jan 2021 – July 2021

- Created PayPal's [DropZone](#) platform's 1st real-time metrics dashboard (JavaScript, React, Google Charts) and created REST APIs (Spring Boot, Elasticsearch), indexing >2M file transfers/hour, adopted globally across 9 regions

PUBLICATIONS

[Google Scholar Link](#)

- [1] Voices from the Margins: Modeling Linguistic Diversity in Spontaneous Speech for Low-Resource Languages, **ACL 2026 (ComputEL workshop)**
- [2] RAINBOW: Reframing Anti LGBTQIA+ Narratives Beyond One World, **In Submission (ARR 2026)**
- [3] On The Conceptualization and Societal Impact of Cross-Cultural Bias, **arXiv Preprint**
- [4] On the Challenges of Building Datasets for Hate Speech Detection, **arXiv Preprint**
- [5] [bitsa_nlp@LT-EDI-ACL2022](#): Leveraging Pretrained Language Models for Detecting Homophobia and Transphobia in Social Media Comments, **ACL 2022 (LT-EDI Workshop)**

SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Technologies: PyTorch, Hugging Face, React, Next.js, Redux, Flask, Oracle SQL, Elasticsearch, Jenkins